

Arnav Vikram Shinde

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☎ 9893183384

📍 Indore, M.P.

🌐 [Arnav Vikram Shinde](#)

Experience

Intern, Gabriel India Ltd.

05/2025 – 07/2025

- Acquired knowledge of suspension systems and damper functionality.
- Studied supply chain management principles and manufacturing line operations.
- Gained expertise in Design Verification Plan (DVP) and Design Failure Mode and Effects Analysis (DFMEA).

Dewas, M.P.

Intern, Avinya MilTech

06/2026 – 07/2026

- Developed precision engineering drawings and 3D models of optical and mechanical components in SolidEdge
- Performed CAE modal, random vibration, and thermal analysis on a satellite-related assembly
- Engineered a military-grade, waterproof enclosure for a Jetson Nano Orin AI camera system, integrating LiDAR, IR, and HD imaging modules
- Contributed to developing a complete kinematic simulator (Creo, MSC ADAMS) for a CMP lens-polishing machine — designing a quick-return-free four-bar oscillating arm whose RPM-induced top disc interacts with a motorized base disc, deriving both discs' profile geometry, computing material removal rates via Preston's equation for production planning, building a predictive re-flattening algorithm to extend disc reusability, and selecting bearings to optimize machine life — resulting in an app that accurately predicts machine behavior.

Indore

Technical Head, Team GSRacers BAJA

03/2025 – 02/2026

Led vehicle design initiatives, coordinated technical aspects, validated designs, and ensured compliance with regulations.

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Virtual Dynamic Head, Team GSRacers BAJA

03/2025 – 02/2026

Modeled virtual model of vehicle on IPG CARMAKER

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Vehicle Dynamics Head, Team GSRacers BAJA

03/2025 – 02/2026

Design Suspension and Steering system for the ATV

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Projects

Reaction Wheel Attitude Control System

02/2026 – 04/2026

- Designed and CAD-modeled a reaction-wheel ACS for a nanosatellite, deriving torque/momentum budgets and moment-of-inertia calculations from rigid-body attitude dynamics
- Built an MBD simulation of the attitude control loop and validated simulated maneuver response against theoretical predictions
- Performed Modal and Random Vibration Analysis in ANSYS to verify structural integrity and natural frequencies under launch loads

Skills

CAD

Creo, SolidWorks, Solidedge, NX CAD

Simulation

ANSYS, MSC Adams, IPG CarMaker

Awards

AIR 1 in Virtual Dynamics (Validation Event) E-BAJA 2026, SAE BAJA

01/2026

AIR 7 overall in M-BAJA 2024, AIR 3 in Physical Dynamics in M-BAJA 202, SAE BAJA

01/2024

Education

Mechanical Eng., *Shri G.S. Institute of Technology And Science*

06/2027
Indore, M.P.

12th, *Vidya Bhavan*

03/2023
Ujjain, M.P.

10th, *Christu Jyotu Convent School*

03/2021
Ujjain, M.P.